



Engineering Chemistry

B.Tech – First Year | Academic Module

UNIT IV

Modern Engineering Materials

Topics Covered

Composites | Refractories | Lubricants | Building Materials

Includes

14 Detailed Sections | SVG Diagrams | Formula Boxes

Q&A Bank | Quick Revision Sheet | Comparison Tables

References

Jain & Jain | Atkins | Taylor | Shaw | Billmeyer

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How to Use This Module: Each section follows a structured format — Definition → Theory → Mechanism → Formulae → Applications. All SVG diagrams are exam-oriented. The Q&A bank is calibrated for 2-mark and 5-mark university questions.

Introduction to Modern Engineering Materials

1.1 What Are Modern Engineering Materials?

Modern engineering materials are substances specifically designed and engineered to meet the demands of advanced technological applications. Unlike conventional materials (iron, wood, stone), modern engineering materials are optimized for combinations of properties such as high strength-to-weight ratio, thermal stability, chemical resistance, and controlled friction behaviour.

The study of modern engineering materials is fundamental to B.Tech because virtually every engineered system — from aircraft fuselages to chemical reactors, from automobile engines to bridge structures — depends on carefully selected materials.

1.2 Scope of Unit IV

Unit IV encompasses four major families of modern engineering materials:

Material Family	Key Concept	Primary Relevance
Composites	Two or more constituent materials combining synergistic properties	Aerospace, construction, automotive
Refractories	High-temperature resistant inorganic materials	Furnaces, kilns, metallurgy
Lubricants	Friction-reducing agents between moving surfaces	Machinery, engines, bearings
Building Materials	Portland cement chemistry and hydration	Civil construction, infrastructure

1.3 Why Study Material Science in Engineering Chemistry?

- Enables selection of appropriate materials for specific engineering tasks.
- Understanding chemical composition helps predict performance and failure modes.
- Optimizing material properties leads to energy efficiency and cost reduction.
- Safety in high-temperature, high-load, or corrosive environments depends on material knowledge.

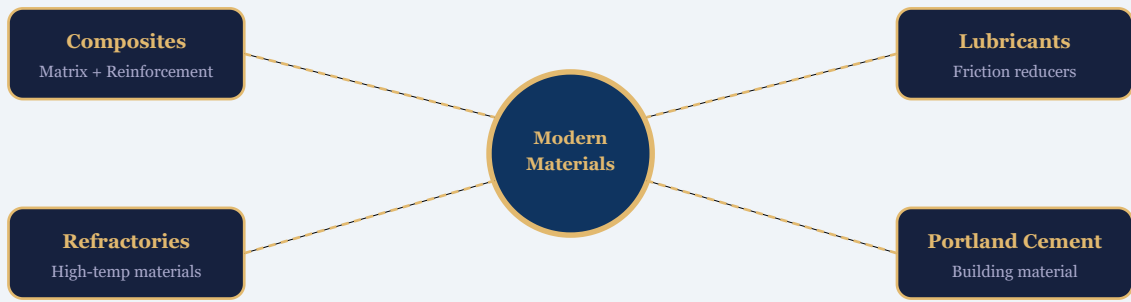


Fig 1.1 – Overview of Modern Engineering Materials Covered in Unit IV

Composites – Basics and Constituents

2.1 Definition of Composites

A **composite material** is a material system composed of two or more physically and/or chemically distinct phases, separated by a distinct interface, combined at a macroscopic level to achieve properties that cannot be obtained by any of the constituent materials acting alone.

Key Point: A composite is NOT a mixture (homogeneous at molecular level) nor an alloy (single-phase after mixing). The phases in a composite remain distinct and identifiable under microscopy.

2.2 Constituents of Composites

Every composite consists of two essential components:

2.2.1 Matrix (Continuous Phase)

The matrix is the primary phase that surrounds and binds the reinforcement. It transfers stress to the reinforcement, protects reinforcement from environmental damage (moisture, oxidation), and gives shape to the composite.

- **Polymer Matrix:** Epoxy, polyester, nylon — lightweight, flexible, easy to process.
- **Metal Matrix:** Aluminium, titanium, copper — high temperature resistance, ductility.
- **Ceramic Matrix:** Silicon carbide, alumina — very high temperature resistance.

2.2.2 Reinforcement (Dispersed Phase)

The reinforcement provides strength, stiffness, and other improved properties to the composite. It is embedded within the matrix in the form of particles, fibres, or layers.

- **Particulate Reinforcement:** Sand, gravel, ceramic particles, metal powders.
- **Fibre Reinforcement:** Glass fibres, carbon fibres, Kevlar fibres, boron fibres.
- **Structural Reinforcement:** Layers, laminates, sandwich panels.

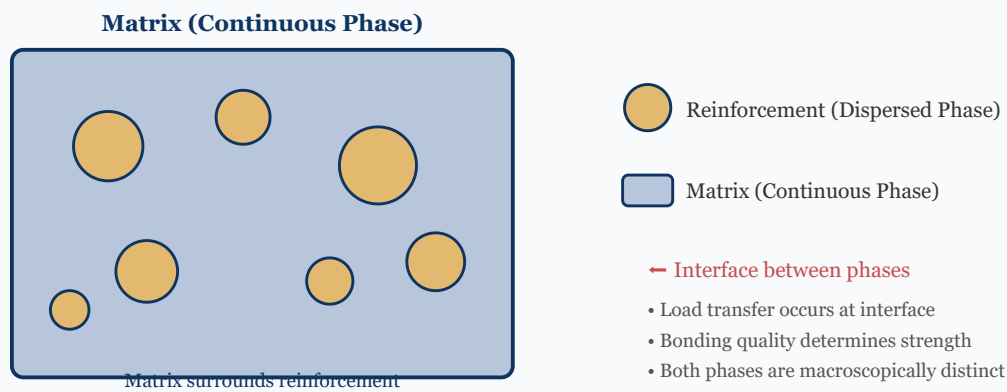


Fig 2.1 – Basic Structure of a Composite Material showing Matrix and Reinforcement Phases

2.3 Interfacial Bonding

The interface between matrix and reinforcement is critical. Good bonding at the interface ensures efficient load transfer from the weaker matrix to the stronger reinforcement. Poor bonding leads to delamination and premature failure.

RULE OF MIXTURES – COMPOSITE PROPERTY ESTIMATION

$$E_c = E_m V_m + E_f V_f$$

where:

E_c = Elastic modulus of composite

E_m = Elastic modulus of matrix | V_m = Volume fraction of matrix

E_f = Elastic modulus of fibre | V_f = Volume fraction of fibre

Note: $V_m + V_f = 1$

Classification of Composites

3.1 Particle Reinforced Composites

Particle reinforced composites contain particles of one material dispersed in a matrix of another. The particles may be spherical, plate-like, or irregular in shape. Particle size generally ranges from 0.01 to 100 μm .

Types:

- **Large Particle Composites:** Interactions with the matrix are not on the atomic/molecular level. Example: Concrete (gravel in cement matrix), WC in Co metal (cutting tools).
- **Dispersion Strengthened Composites:** Extremely small particles (10–100 nm) impede dislocation movement. Example: Thoria-dispersed (TD) nickel, sintered aluminium powder (SAP).

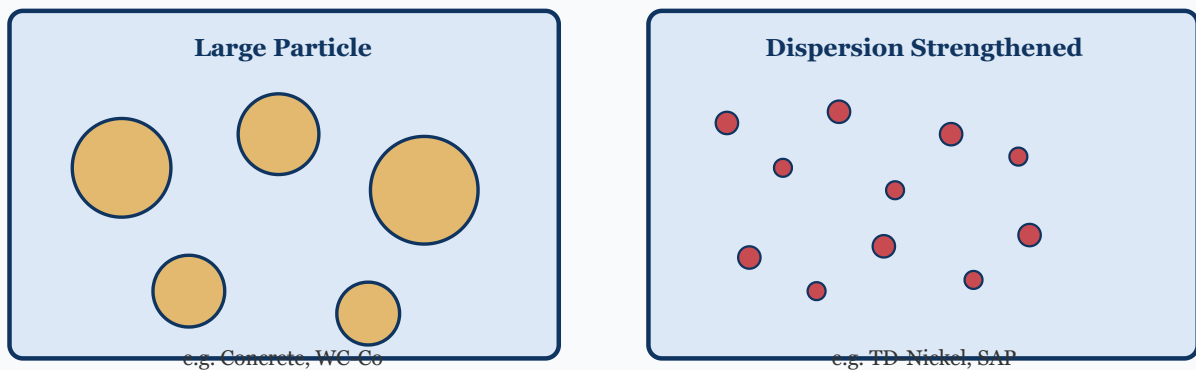


Fig 3.1 – Particle Reinforced Composites: Large Particle vs Dispersion Strengthened

3.2 Fibre Reinforced Composites

Fibre reinforced composites (FRC) use fibres as the primary reinforcement embedded in a matrix. Fibres provide high strength and stiffness along their length, making them the most effective form of reinforcement.

Types of Fibre Reinforcement:

- **Continuous (aligned) fibres:** Long fibres aligned in one direction — anisotropic, very strong along fibre axis.

- **Discontinuous (chopped) fibres:** Short fibres randomly or aligned — more isotropic but lower strength.
- **Woven fibres:** Fibres woven into a fabric before embedding — balanced properties in two directions.

Common fibres used: Glass fibre (GFRP), Carbon fibre (CFRP), Kevlar (aramid), Boron fibre.

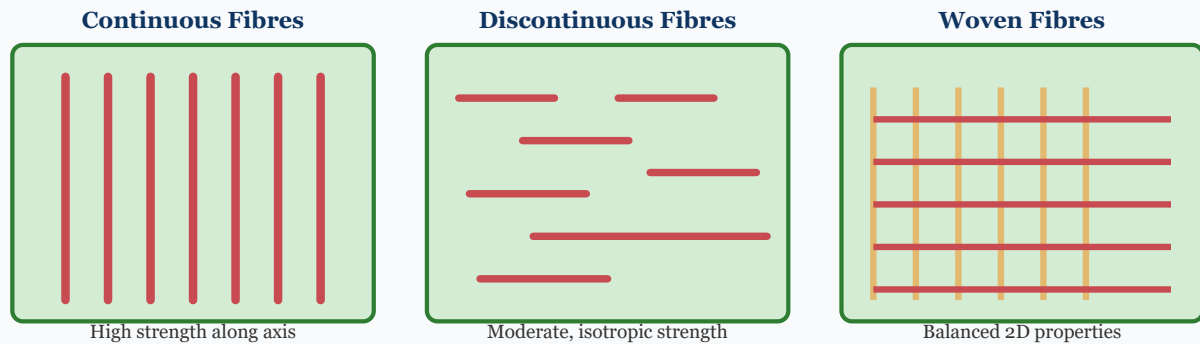


Fig 3.2 – Types of Fibre Reinforcement in Composites

3.3 Structural Composites

Structural composites are formed by combining different materials in a layered or spatial arrangement where the geometry of the arrangement is the key design variable.

Types:

- **Laminates:** Layers of fibrous composite sheets bonded together with varying fibre orientations. Example: Plywood, fibre glass laminates.
- **Sandwich Panels:** Two stiff, strong face sheets (skins) bonded to a low-density core material (foam, honeycomb). Used in aircraft, building panels.

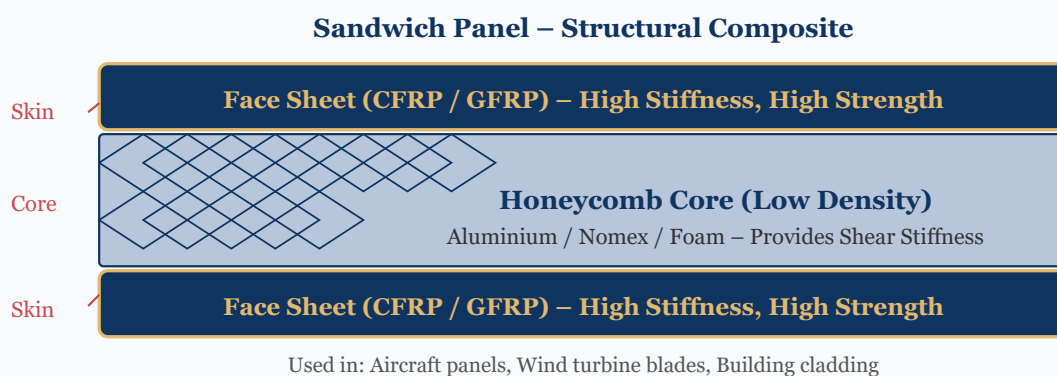


Fig 3.3 – Sandwich Panel Structural Composite showing Face Sheets and Core

Properties and Applications of Composites

4.1 Properties of Composite Materials

Property	Characteristic	Compared to Metals
Strength-to-weight ratio	Very high	Up to 5× better
Stiffness (Elastic Modulus)	High (especially CFRP)	Carbon fibre 3–4× stiffer per unit mass
Fatigue resistance	Excellent	Cracks do not propagate easily across fibres
Corrosion resistance	Excellent (polymer matrix)	No rust, no galvanic corrosion
Thermal conductivity	Generally low	Good thermal insulator
Design flexibility	Very high	Tailorable anisotropic properties
Cost	Higher	More expensive to manufacture

4.2 Applications of Composites

Aerospace and Defence

CFRP used in Boeing 787 Dreamliner (50% by weight), Airbus A350 fuselage, military aircraft wings, helicopter rotor blades, missile bodies. Reduces weight → fuel efficiency → longer range.

Automotive

GFRP body panels, CFRP chassis in Formula 1, metal matrix composites (MMC) for brake discs. Reduces vehicle weight → better fuel economy.

Civil Engineering and Construction

GFRP rebar in concrete (non-corrosive substitute for steel), fibre reinforced concrete, composite bridge decks, FRP wrapping for column strengthening.

Sports and Consumer Products

Carbon fibre bicycle frames, tennis rackets, golf clubs, ski poles. High stiffness + low weight = performance enhancement.

Marine Industry

GFRP boat hulls, ship superstructures, submarine fairings — excellent corrosion resistance in saline environments.

Key Comparison – GFRP vs CFRP vs KFRP:

GFRP (Glass) – Low cost, moderate strength, good chemical resistance. Used in boats, pipes.

CFRP (Carbon) – Very high strength/stiffness, expensive, used in aerospace.

KFRP (Kevlar/Aramid) – Very high toughness, impact resistant. Used in body armour, helmets.

4.3 Advantages and Limitations of Composites

Advantages	Limitations
High specific strength and stiffness	High manufacturing cost
Excellent fatigue and corrosion resistance	Difficult to repair after damage
Tailorable anisotropic properties	Delamination under transverse loads
Near-net shape manufacturing possible	Recycling is complex
Design flexibility and weight reduction	Quality control requires NDT

Refractories – Introduction and Classification

5.1 Definition of Refractories

A **refractory material** is a material that retains its strength and chemical stability at very high temperatures (above 1580°C) and is therefore used in the construction of high-temperature industrial equipment such as furnaces, kilns, incinerators, and reactors.

Definition (ASTM): "Non-metallic materials having those chemical and physical properties that make them applicable for structures, or as components of systems, that are exposed to environments above 1000°F (538°C)."

5.2 Classification of Refractories

5.2.1 Based on Chemical Composition

- **Acidic Refractories:** Contain acidic oxides (SiO_2 , Al_2O_3 <72%). Resist attack by acidic slags. Examples: Silica brick, fireclay brick, aluminosilicate brick.
- **Basic Refractories:** Contain basic oxides (MgO , CaO). Resist basic slags. Examples: Magnesite brick, dolomite brick, chrome-magnesite brick.
- **Neutral Refractories:** Contain neutral oxides. Resist both acidic and basic slags. Examples: Chromite brick, carbon block, silicon carbide.

5.2.2 Based on Fusion Temperature (Refractoriness)

Class	PCE (Pyrometric Cone Equivalent)	Temperature Range
Low Heat Duty	PCE 15–19	1520–1630°C
Medium Heat Duty	PCE 19–26	1520–1630°C
High Heat Duty	PCE 26–33	1630–1750°C
Super Duty	PCE 33–38	>1750°C

5.2.3 Based on Physical Form

- **Shaped Refractories:** Bricks, tiles, special shapes — pre-fired and ready to use.
- **Unshaped (Monolithic) Refractories:** Castables, ramming mixes, gunning mixes, plastic mixes — applied wet and fired in situ.

- **Refractory Fibres:** Ceramic fibre blankets, modules — used for furnace insulation lining.

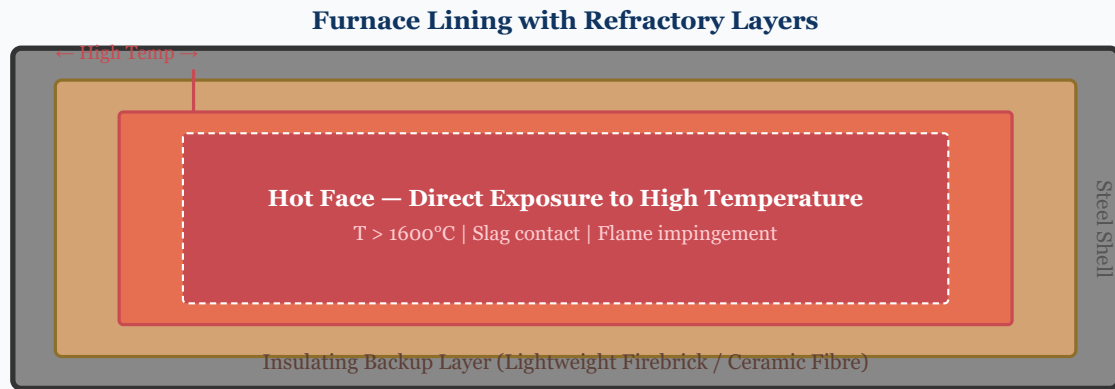


Fig 5.1 – Cross-section of Furnace Lining with Refractory Layers

REFRACTORINESS — PCE DETERMINATION

PCE (Pyrometric Cone Equivalent): Measured by comparing softening of refractory sample with standard Orton/Segger cones at heating rate of 5°C/min.

Refractoriness Under Load (RUL): Temperature at which refractory deforms by 0.5% under a compressive load of 0.2 N/mm²

Properties of Refractories

6.1 Refractoriness (PCE)

Refractoriness is the ability of a refractory material to withstand high temperatures without softening, melting, or deforming under its own weight. It is measured by the Pyrometric Cone Equivalent (PCE) – the higher the PCE number, the greater the refractoriness.

6.2 Refractoriness Under Load (RUL)

In actual service, refractories bear structural loads at high temperatures. RUL measures the temperature at which a standard test specimen (50mm cube) deforms by 0.5% (T_{0.5}) and 10% (T₁₀) under a standard load of 0.2 N/mm². A high RUL indicates better structural integrity.

6.3 Thermal Spalling Resistance

Spalling is the cracking and breaking of refractory bricks due to rapid temperature changes (thermal shock). It occurs due to differential thermal expansion. Good refractories have low thermal expansion coefficient and high thermal conductivity to minimise thermal gradients.

6.4 Chemical Properties

- **Slag Resistance:** Ability to resist attack by acidic or basic slags in contact with the hot face.
- **Oxidation Resistance:** Resistance to oxidative atmospheres at high temperature.
- **Chemical Inertness:** Should not react with the furnace atmosphere or the material being processed.

6.5 Physical Properties

Property	Significance	Typical Values
Apparent Porosity (%)	Lower porosity → better slag and gas penetration resistance	10–25%
Bulk Density (g/cm ³)	Higher density → greater strength	1.8–3.0
Cold Crushing Strength (CCS)	Ability to bear load at room temperature	10–200 N/mm ²

Permanent Linear Change (PLC)	Shrinkage or expansion on repeated heating	<±2%
Thermal Conductivity (W/mK)	Controls heat losses; low → better insulation	0.5–30

6.6 Thermal Properties

- **Thermal Conductivity:** Should be low for furnace walls (to conserve heat) but controlled for hearths (to remove heat).
- **Thermal Expansion:** Should be low and uniform to prevent spalling.
- **Heat Capacity:** High heat capacity slows temperature fluctuations.

Properties of Refractories – Spider Chart (Relative Comparison)

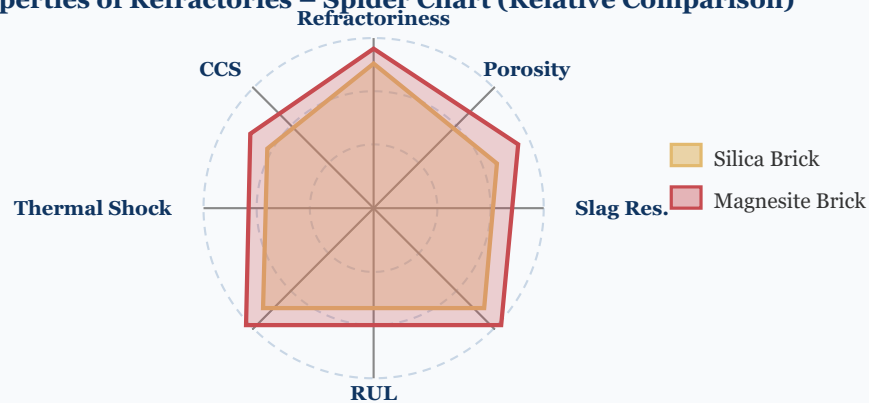


Fig 6.1 – Relative Property Comparison of Common Refractory Types

Factors Affecting Refractory Materials

7.1 Temperature

Temperature is the most critical factor. At high temperatures refractories may undergo phase transformations, softening, and creep. The operating temperature must remain below the refractory's RUL. Temperature fluctuations cause thermal spalling.

7.2 Load / Stress

Refractories in furnace roofs, arches, and walls bear significant structural loads. High load combined with high temperature causes creep (slow plastic deformation) leading to eventual collapse. RUL and creep resistance must match service conditions.

7.3 Chemical Attack

- **Slag Attack:** Molten slag dissolves the refractory surface. Acidic slags attack basic refractories; basic slags attack acidic refractories.
- **Gaseous Attack:** CO, SO₂, H₂O vapour can react with refractories — CO causes carbon deposition in pores.
- **Metal Penetration:** Molten metals can penetrate porous refractories, weakening structure.

7.4 Thermal Shock

Sudden temperature changes cause differential expansion → cracking. Resistance to thermal shock depends on: low coefficient of thermal expansion, high thermal conductivity, high strength, and low elastic modulus.

THERMAL SHOCK RESISTANCE PARAMETER (R)

$$R = (\sigma_f \times k) / (E \times \alpha)$$

where:

σ_f = flexural strength (MPa)

k = thermal conductivity (W/mK)

E = elastic modulus (GPa)

alpha = coefficient of thermal expansion (per °C)
Higher R → better thermal shock resistance

7.5 Abrasion and Erosion

In cement kilns, blast furnaces, and similar applications, refractories face abrasion from solid charge materials and erosion from high-velocity gases carrying particulates. Dense, fine-grained refractories with low porosity offer better abrasion resistance.

7.6 Porosity and Permeability

High open porosity allows molten metal, slag, and gases to penetrate the refractory, causing chemical attack from within. Low porosity (<15%) is generally desirable. However, some porosity allows thermal shock accommodation.

7.7 Mineralogical Composition

The phase composition of a refractory determines its refractoriness. For example, silica brick with >95% SiO₂ exists as cristobalite and tridymite — high temperature stable phases. Presence of impurities (K₂O, Na₂O, FeO) lowers the fusion point by forming low-melting eutectics.

Factor	Effect on Refractory	Mitigation
High Temperature	Creep, softening, phase change	Select higher PCE material
Thermal Shock	Cracking, spalling	Controlled heating, fibre-reinforced refractories
Slag Attack	Dissolution of surface	Match chemistry (acid/basic/neutral)
Load	Creep deformation	High RUL material, proper support design
Abrasion	Surface wear	Dense, low-porosity castables
Impurities	Lowered fusion point	Use high-purity raw materials

Applications of Refractories

8.1 Iron and Steel Industry

The largest consumer of refractories. Used in blast furnaces (fireclay and high-alumina bricks), BOF (basic oxygen furnace) converters (magnesite and dolomite bricks), electric arc furnaces, ladles, tundishes, and continuous casting equipment.

8.2 Cement Industry

Rotary kilns for clinker production use magnesite-chrome and dolomite refractories in the burning zone (1400–1500°C). Cooler area uses high-alumina castables. Cyclone preheaters use aluminosilicate castables.

8.3 Glass Industry

Glass melting furnaces (1400–1600°C) use fused cast AZS (Alumina-Zirconia-Silica) blocks for the glass contact zone, silica crown bricks for the furnace crown, and insulating firebricks for the backup.

8.4 Non-Ferrous Metal Industry

Copper smelting, aluminium smelting, and zinc refining use acid-resistant and basic refractories depending on the slag chemistry.

8.5 Petrochemical Industry

Hydrocarbon cracking furnaces, reformers, and heaters use castable refractories, ceramic fibre modules, and gunning mixes for maintenance.

8.6 Household and Small Industrial Uses

Kiln furniture for pottery/ceramics, muffle furnaces in laboratories, pizza ovens and tandoors (fireclay bricks), incinerators (high-alumina castables).

Application	Type of Refractory	Key Requirement
Blast Furnace Hearth	Carbon blocks	Conductivity + chemical resistance to iron

Ladle Lining	Magnesia-carbon bricks	Basic slag resistance
Glass Furnace Crown	Fused silica bricks	Low creep, low expansion
Cement Kiln Burning Zone	Magnesia-spinel bricks	High RUL, thermal shock resistance
Petrochemical Heater	Ceramic fibre modules	Low thermal conductivity (insulation)
Incinerator	High-alumina castable	Chemical attack resistance

Industry Insight: Refractories are "consumable" materials — they erode slowly with each campaign. The economic service life of a refractory lining, measured in tonnes of steel or cement produced per unit of refractory consumed, is a key industry metric. Modern refractories last 10–25 years in cement kilns.

Lubricants – Classification and Functions

9.1 Definition of Lubricants

A **lubricant** is a substance (solid, liquid, semisolid, or gaseous) introduced between two surfaces in relative motion to reduce friction and wear, prevent damage to the surfaces, and carry away heat generated during motion.

9.2 Functions of Lubricants

- **Reduce Friction:** Primary function — minimise resistance to motion between sliding/rolling surfaces.
- **Reduce Wear:** Prevent metal-to-metal contact which causes material removal.
- **Cooling:** Carry away frictional heat, especially in engine oils (lubricant circulates).
- **Sealing:** Grease acts as a seal to prevent entry of contaminants (dust, water).
- **Corrosion Protection:** Lubricant film prevents moisture and oxygen contact with metal surfaces.
- **Cleaning:** Detergent additives in engine oils suspend and carry away soot, combustion deposits.
- **Shock Absorption:** Lubricant film cushions impact loads in gear systems.
- **Power Transmission:** Hydraulic fluids transmit power in hydraulic systems.

9.3 Classification of Lubricants

9.3.1 Based on Physical State

Type	Examples	Applications
Liquid Lubricants (Oils)	Mineral oil, synthetic oil, vegetable oil	Engines, turbines, machine tools
Semi-solid (Greases)	Lithium grease, calcium grease, sodium grease	Wheel bearings, slow-speed machinery
Solid Lubricants	Graphite, MoS ₂ , PTFE (Teflon), talc	Extreme temperature, vacuum applications

Gaseous Lubricants	Air, nitrogen	Gas bearings, dental drills, high-speed spindles
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9.3.2 Based on Origin

- **Animal and Vegetable Oils:** Castor oil, lard oil, whale oil — biodegradable, good lubricity but poor oxidation stability.
- **Mineral Oils (Petroleum-based):** Derived from crude oil by distillation and refining. Most widely used — paraffinic, naphthenic, aromatic types.
- **Synthetic Lubricants:** Polyalphaolefins (PAO), esters, polyglycols, silicone oils — superior performance, wider temperature range.

9.3.3 Based on Mechanism of Action

- Thick-film (hydrodynamic) lubricants
- Thin-film (boundary) lubricants
- Extreme Pressure (EP) lubricants

9.4 Lubricant Additives

Additive Type	Function	Example
Anti-oxidants	Retard oil oxidation	Phenolic antioxidants, ZDDP
Viscosity Index Improvers	Reduce viscosity change with temperature	Polymethacrylates
Anti-wear agents	Protect surfaces under mixed film conditions	ZDDP, tricresyl phosphate
Extreme Pressure (EP) agents	React with metal at asperities to form protective layer	Sulphur, chlorinated paraffins
Detergents	Keep engine clean, neutralise acids	Calcium sulphonate
Pour Point Depressants	Lower the pour point	Alkylnaphthalene

Mechanism of Lubrication

10.1 Thick Film (Hydrodynamic) Lubrication

In hydrodynamic (thick film) lubrication, a continuous, unbroken film of lubricant completely separates the two surfaces. There is NO metal-to-metal contact. The film is maintained by the relative motion of the surfaces which generates a hydrodynamic pressure (wedge action) sufficient to support the load.

Characteristics: Very low friction (coefficient ≈ 0.001 – 0.01), dependent on oil viscosity, operates at high speeds and light-to-moderate loads, friction governed by viscous shear of lubricant.



10.2 Thin Film (Boundary) Lubrication

In boundary lubrication, the load is so high or the speed so low that the lubricant film becomes very thin (a few molecular layers), and some metal-to-metal contact occurs at surface asperities. The lubricant molecules are physically or chemically adsorbed onto the metal surface, forming a protective layer that reduces friction and wear at contact points.

Characteristics: Friction coefficient ≈ 0.05 – 0.15 , depends on chemical nature of lubricant (not viscosity), operates at low speeds/high loads, critical for start-stop cycles.

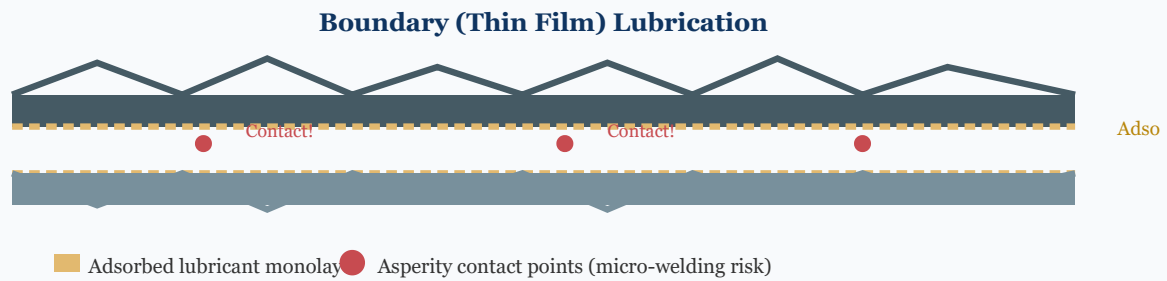


Fig 10.2 – Boundary Lubrication showing Adsorbed Film and Asperity Contacts

10.3 Extreme Pressure (EP) Lubrication

When loads are very high and boundary lubrication is insufficient, EP lubricants containing sulphur, chlorine, or phosphorus compounds are used. At the high temperatures generated at asperity contacts, these compounds react with the metal surface to form iron sulphide, iron chloride, or iron phosphate — thin, solid, low-shear-strength compounds that prevent seizure (welding together of surfaces).

STRIBECK CURVE – LUBRICATION REGIMES

Friction Coefficient (μ) vs. Bearing Number ($\eta \times N / P$)

η = viscosity | N = speed (rpm) | P = pressure (N/m^2)

Low $\eta \times N / P \rightarrow$ Boundary lubrication (high μ , $\sim 0.1-0.5$)

Medium $\eta \times N / P \rightarrow$ Mixed film (transition)

High $\eta \times N / P \rightarrow$ Hydrodynamic lubrication (low μ , $\sim 0.001-0.01$)

Regime	Film Thickness	Friction Coefficient	Controlling Factor
Hydrodynamic	10–100 μm	0.001–0.01	Oil viscosity
Elastohydrodynamic	0.1–1 μm	0.01–0.1	Elastic deformation + viscosity
Boundary	1–10 nm (molecular)	0.05–0.15	Surface chemistry
Dry / No lubrication	0	0.2–0.5+	Surface roughness, material

Properties of Lubricating Oils

11.1 Viscosity

Definition: Viscosity is the resistance of a fluid to flow (internal friction). It is the most important property of a lubricating oil. A highly viscous oil flows slowly; a low viscosity oil flows freely.

- **Dynamic (Absolute) Viscosity (η):** Force per unit area required to maintain a unit velocity gradient. Unit: Pa·s or poise (P).
- **Kinematic Viscosity (ν):** Dynamic viscosity divided by density. Unit: m²/s or Stoke (St), centistoke (cSt).

VISCOSITY DEFINITIONS

Dynamic Viscosity: $\eta = \text{Shear Stress} / \text{Shear Rate} = (F/A) / (dv/dy)$
[Units: Pa.s]

Kinematic Viscosity: $\nu = \eta / \rho$ [Units: m²/s or cSt]

where ρ = density of oil (kg/m³)

Effect of Temperature on Viscosity: Viscosity decreases with increase in temperature. All lubricating oils show this behaviour. The rate of change varies — a good lubricating oil should show minimal change in viscosity with temperature (high Viscosity Index).

Effect of Pressure on Viscosity: Viscosity increases with increase in pressure (important in elastohydrodynamic lubrication).

11.2 Viscosity Index (VI)

Definition: Viscosity Index is an arbitrary, dimensionless number indicating the change in viscosity of an oil with temperature. A HIGH viscosity index means the oil's viscosity changes LESS with temperature — which is desirable.

VISCOSITY INDEX FORMULA (DEAN AND DAVIS)

$$VI = (L - U) / (L - H) \times 100$$

where:

U = kinematic viscosity of oil at 40°C (sample)

L = viscosity at 40°C of a reference oil (VI = 0) having same viscosity at 100°C as sample

H = viscosity at 40°C of a reference oil (VI = 100) having same viscosity at 100°C as sample

Higher VI → Better viscosity-temperature characteristics

VI > 100 → Multigrade oils (e.g., 10W-40)

Viscosity vs Temperature – Effect of Viscosity Index

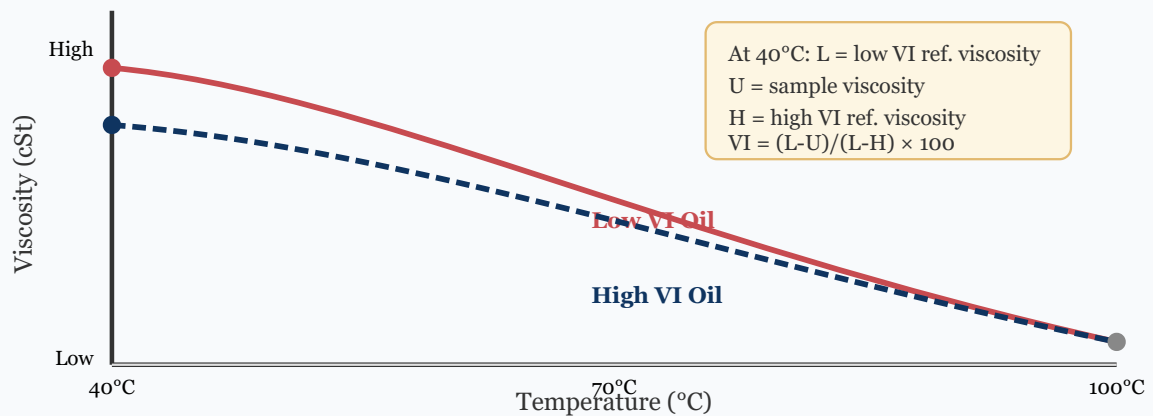


Fig 11.1 – Viscosity vs Temperature showing Low VI vs High VI Lubricating Oils

11.3 Flash Point and Fire Point

Flash Point: The lowest temperature at which an oil gives off sufficient vapour to form a momentarily flammable mixture with air when a small flame is applied. The vapour ignites but does not sustain combustion.

Fire Point: The temperature at which the oil vapour continues to burn for at least 5 seconds after ignition. Fire point is always higher than flash point (typically 10–30°C above).

FLASH & FIRE POINT SIGNIFICANCE

Flash Point → Safety indicator (minimum temperature for safe storage and handling)

Fire Point → Fire hazard indicator (minimum temperature for sustained combustion)

Typical values:

Mineral oil: Flash Point 180–240°C | Fire Point 200–280°C

Engine oil SAE 40: Flash Point ~220°C
Higher Flash Point → Safer lubricant

11.4 Cloud Point and Pour Point

Cloud Point: The temperature at which a lubricating oil becomes hazy or cloudy due to the separation of wax (paraffin) crystals. The oil starts to lose clarity and flow characteristics begin to deteriorate.

Pour Point: The temperature at which the oil just ceases to flow (3°C above the temperature at which flow stops). Below the pour point, the oil is too viscous or solid to flow and cannot lubricate properly.

The pour point is typically 3–6°C below the cloud point. Both are important for cold-climate applications (arctic lubricants, aviation hydraulics).

11.5 Saponification Value

Definition: The number of milligrams of KOH required to saponify (hydrolyse and neutralise) 1 gram of oil. It indicates the degree of unsaturation and presence of ester-based components in the oil.

SAPONIFICATION NUMBER

Saponification Value (SV) = mg of KOH required to saponify 1 g of oil

High SV → More ester/animal/vegetable oil content → Better lubricity at boundary conditions

Mineral oils: SV near zero (no esters)

Lard oil / castor oil: SV = 180–200 mg KOH/g

Property	Significance	Desirable Value
Viscosity	Film forming ability, flow resistance	Appropriate for application
Viscosity Index	Viscosity change with temperature	High (>100 for multigrade)
Flash Point	Fire safety	High (>180°C)
Fire Point	Sustained fire hazard	High (>200°C)
Cloud Point	Wax separation temperature	Low (for cold climates)
Pour Point	Low temperature flowability	Low (for cold climates)

Saponification Value	Ester content, lubricity	High for boundary conditions
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Building Materials – Portland Cement

12.1 Introduction to Portland Cement

Portland cement is the most widely used hydraulic binder in construction. It is a **hydraulic cement**, meaning it sets and hardens by chemical reaction with water (hydration) and retains strength even underwater. It was developed by Joseph Aspdin (England, 1824) and named after Portland stone which it resembles in colour and quality.

12.2 Raw Materials for Portland Cement

- **Calcareous materials:** Limestone (CaCO_3), chalk, marl — provide calcium oxide (CaO).
- **Argillaceous materials:** Clay, shale — provide silica (SiO_2), alumina (Al_2O_3), iron oxide (Fe_2O_3).
- **Minor additions:** Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) — added to regulate setting time.

12.3 Constituents (Compounds) of Portland Cement

After burning (clinkering) at $1400\text{--}1500^\circ\text{C}$, the raw materials form four main clinker compounds (Bogue compounds):

Compound	Chemical Name	Formula	Abbrev.	Content (%)	Role
Tricalcium Silicate	Alite	$3\text{CaO} \cdot \text{SiO}_2$	C_3S	45–65%	Early strength, heat liberation
Dicalcium Silicate	Belite	$2\text{CaO} \cdot \text{SiO}_2$	C_2S	15–30%	Long-term strength
Tricalcium Aluminate	Celite	$3\text{CaO} \cdot \text{Al}_2\text{O}_3$	C_3A	6–12%	Flash setting, early heat
Tetracalcium Aluminoferrite	Ferrite	$4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$	C_4AF	6–10%	Colour, moderate heat

Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, 2–4%) is added to the clinker during grinding to control the flash-setting tendency of C_3A by forming ettringite (calcium sulphoaluminate).

Portland Cement Manufacturing – Dry Process Flowchart



Fig 12.1 – Portland Cement Manufacturing Process (Dry Process)

12.4 Properties of Good Cement

- Initial setting time: minimum 30 minutes (IS: 269)
- Final setting time: maximum 600 minutes (10 hours)
- Fineness: Specific surface area $\geq 225 \text{ m}^2/\text{kg}$ (controls hydration rate)
- Soundness: No excessive expansion after setting (Le-Chatelier test)
- Compressive strength: $\geq 33, 43, \text{ or } 53 \text{ MPa}$ at 28 days (grade dependent)

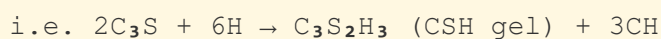
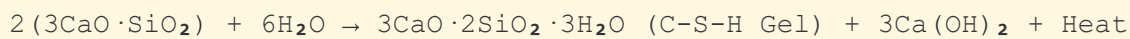
Setting and Hardening of Cement

13.1 Hydration of Cement

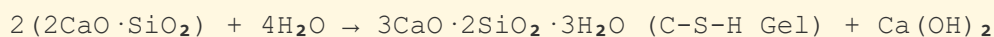
When water is added to cement, the clinker compounds undergo **hydration** — chemical reactions with water that produce hydrated compounds responsible for strength. This is an exothermic process (heat of hydration).

13.1.1 Hydration Reactions

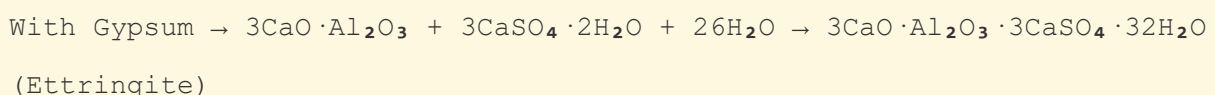
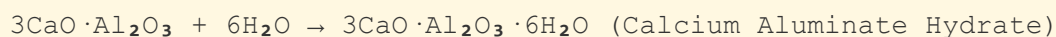
C₃S Hydration (Alite — Fast, Early Strength) :



C₂S Hydration (Belite — Slow, Long-term Strength) :

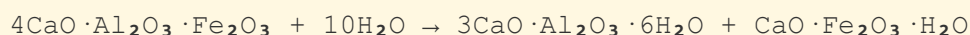


C₃A Hydration (Celite — Very Fast, Controls Flash Setting) :



Ettringite controls early stiffening — prevents flash set

C₄AF Hydration (Ferrite — Moderate Rate) :



13.2 Setting of Cement

Setting refers to the stiffening and solidification of the cement paste. It occurs in two stages:

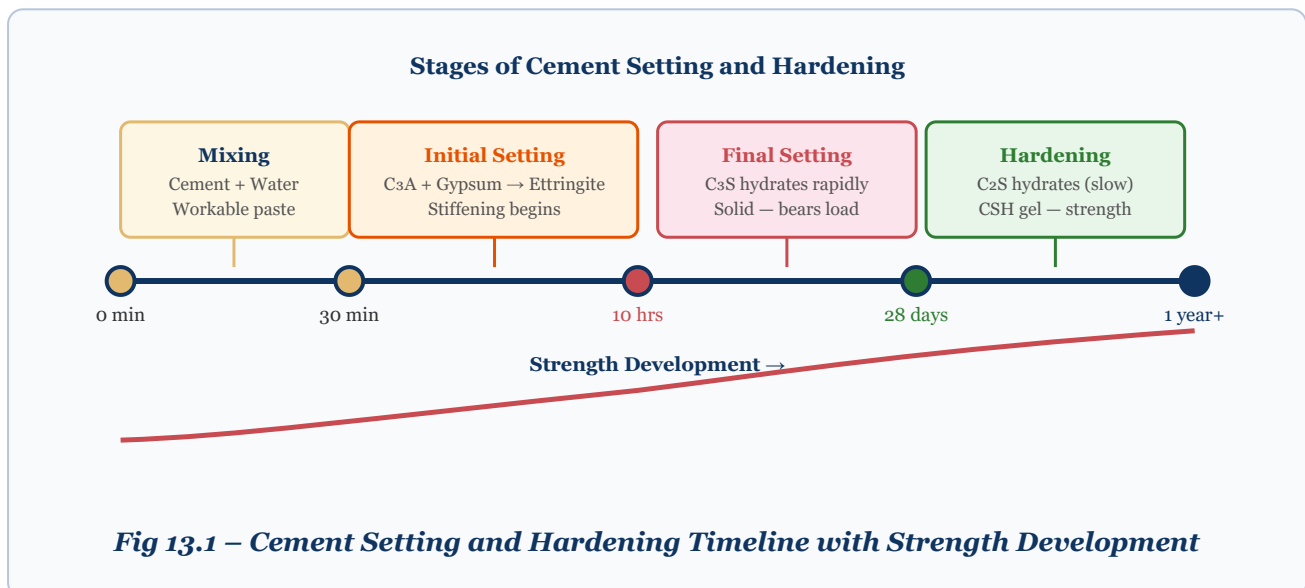
- **Initial Setting (≥ 30 min):** Cement paste begins to stiffen and lose plasticity. The paste can no longer be moulded. This is primarily due to rapid hydration of C₃A forming ettringite.
- **Final Setting (≤ 10 hours):** Cement paste completely solidifies and can bear a small load without imprint. All surface hydration is essentially complete.

Important: Concrete must be placed and compacted before initial set occurs (usually within 45–60 minutes of mixing). If disturbed after initial set, the concrete loses strength — a phenomenon called "retempering damage."

13.3 Hardening of Cement

Hardening is the process by which the set cement gains strength over time. It continues for months and years. Hardening occurs due to:

- Continued hydration of C_3S and C_2S producing C-S-H gel which fills pore spaces.
- The C-S-H gel (calcium silicate hydrate) is the primary strength-giving phase — a complex, poorly crystalline material with high surface area ($150\text{--}700\text{ m}^2/\text{g}$).
- $\text{Ca}(\text{OH})_2$ (portlandite) crystals fill remaining pore space but are not strong — can react with CO_2 (carbonation) or sulphates (sulphate attack).



13.4 Heat of Hydration

Cement hydration is exothermic. The heat generated (heat of hydration) must be controlled in mass concrete structures to prevent thermal cracking. C_3A and C_3S generate the most heat. Low-heat Portland cement (LHPC) is used in dams and large foundations — it has lower C_3A and C_3S content.

HEAT OF HYDRATION (APPROXIMATE VALUES)

$C_3S \rightarrow 502\text{ J/g}$

$C_2S \rightarrow 260\text{ J/g}$

$C_3A \rightarrow 867\text{ J/g}$ (HIGHEST — main cause of flash set)

$C_4AF \rightarrow 419\text{ J/g}$

Total heat for OPC (Ordinary Portland Cement) $\approx$ 350–400 J/g at 7 days

13.5 Setting vs Hardening – Comparison

Parameter	Setting	Hardening
Definition	Stiffening and solidification of paste	Gain of strength by set paste
Time scale	30 min – 10 hours	Days, weeks, months, years
Primary compounds	C ₃ A, C ₃ S (early)	C ₃ S, C ₂ S (continuing)
Product formed	Ettringite, initial C-S-H	C-S-H gel, Ca(OH) ₂
Physical change	Loss of plasticity	Increase in compressive strength
Water requirement	Water/cement ratio critical	Curing water essential

Questions & Answers

Part A – 2 Mark Questions

Q 1

2 Marks

Define a composite material and name its constituents.

A composite material is a material system consisting of two or more physically and chemically distinct phases, separated by a distinct interface, combined macroscopically to achieve properties superior to those of individual components acting alone. The two essential constituents are: (1) the **Matrix** — the continuous phase that binds and surrounds the reinforcement, transfers load, and provides shape; and (2) the **Reinforcement** — the dispersed phase (particles, fibres, or layers) that provides enhanced strength, stiffness, or other properties. The matrix can be polymeric, metallic, or ceramic.

Q 2

2 Marks

Define refractories. What is PCE?

Refractories are non-metallic materials that retain their strength and chemical inertness at very high temperatures (above 1580°C) and are used in the construction of furnaces, kilns, reactors, and high-temperature industrial equipment. They resist thermal, chemical, and physical stresses at elevated temperatures. **PCE (Pyrometric Cone Equivalent)** is the measure of refractoriness — the cone number of the standard Orton/Seeger cone that bends at the same temperature as the refractory test piece under standard heating conditions. Higher PCE number indicates higher refractoriness (greater ability to withstand temperature).

Q 3

2 Marks

What is viscosity index? Give its formula.

Viscosity Index (VI) is an arbitrary, dimensionless number that indicates the degree of change in viscosity of a lubricating oil with temperature. A high VI means the oil's viscosity changes very little with temperature change, which is highly desirable for engine and gear oils that operate across a wide temperature range. The formula is: $VI = (L - U) / (L - H) \times 100$, where U is the kinematic viscosity of the sample oil at 40°C; L is the viscosity at 40°C of a reference oil with VI = 0, having same viscosity as the sample at 100°C; H is the viscosity at 40°C of a reference oil with VI = 100, having same viscosity as sample at 100°C. Multigrade oils (e.g., 10W-40) have VI > 100.

Q 4**2 Marks****What are the Bogue compounds in Portland cement?**

The four major clinker compounds (Bogue compounds) in Portland cement are: (1) **Tricalcium Silicate (C_3S , Alite)** — 45–65%, responsible for early strength and high heat of hydration; (2) **Dicalcium Silicate (C_2S , Belite)** — 15–30%, responsible for slow long-term strength gain; (3) **Tricalcium Aluminate (C_3A , Celite)** — 6–12%, causes flash setting, generates maximum heat; controlled by gypsum addition; (4) **Tetracalcium Aluminoferrite (C_4AF , Ferrite)** — 6–10%, gives grey colour, moderate heat. Additionally, **Gypsum** (2–4%) is added to control flash setting by reacting with C_3A to form ettringite.

Q 5**2 Marks****Distinguish between flash point and fire point.**

Flash Point is the lowest temperature at which a lubricating oil produces sufficient vapour to form a momentarily flammable mixture with air upon exposure to a small ignition source. The vapour ignites briefly but does not sustain combustion — it "flashes." **Fire Point** is the temperature at which the oil vapour ignites and continues to burn for at least 5 seconds after ignition. The fire point is always 10–30°C higher than the flash point. Flash point indicates safe handling/storage temperature, while fire point indicates the temperature of sustained fire hazard. For a good lubricating oil, both should be as high as possible (flash point > 180°C for mineral oils).

Q 6**2 Marks****What is meant by setting and hardening of cement?**

Setting of cement refers to the process by which fresh cement paste stiffens and solidifies, losing its plasticity and workability. It occurs in two stages: (i) Initial Setting — paste begins to stiffen (min. 30 min from mixing) due to C_3A hydration forming ettringite; (ii) Final Setting — paste completely solidifies (max. 600 min). **Hardening** refers to the subsequent development of compressive strength in the solidified cement paste over days, weeks, and months. It occurs due to continued hydration of C_3S and C_2S , forming C-S-H gel which fills pore spaces and bonds aggregate particles. Setting is primarily a physical change (solidification), while hardening is a physico-chemical process involving progressive strength gain.

Part B – 5 Mark Questions

Explain the classification of composite materials with examples.

Composite materials are classified into three major types based on the nature of reinforcement:

(A) Particle Reinforced Composites: Reinforced with particles of a second phase dispersed in the matrix. Classified into: (i) Large Particle Composites — particle size $>1\ \mu\text{m}$, interaction not molecular, e.g., concrete (gravel in cement), WC-Co cutting tools, road asphalt; (ii) Dispersion Strengthened Composites — ultra-fine particles (10–100 nm) impede dislocation motion, e.g., TD-Nickel (ThO_2 in Ni), sintered aluminium powder (SAP). Properties are relatively isotropic since particles can be randomly oriented.

(B) Fibre Reinforced Composites (FRC): Reinforced with fibres (continuous or discontinuous) providing high strength and stiffness along fibre direction. Types: (i) Continuous (aligned) fibre — high anisotropic strength, e.g., CFRP in aircraft; (ii) Discontinuous (random) fibre — moderate isotropic strength, e.g., chopped glass fibre in automotive panels; (iii) Woven fibre — balanced 2D properties, e.g., glass fabric composites. The rule of mixtures applies: $E_c = E_m V_m + E_f V_f$. Key fibres: Glass (GFRP), Carbon (CFRP), Kevlar (KFRP).

(C) Structural Composites: The geometry/arrangement of constituents determines properties. Types: (i) Laminates — layers of fibrous composites stacked with varying fibre orientations for quasi-isotropic behaviour, e.g., plywood, fibre-glass laminates; (ii) Sandwich Panels — stiff, strong face sheets (CFRP, GFRP) bonded to lightweight, low-density core (honeycomb aluminium, Nomex foam) — very high flexural stiffness with low weight. Used in aircraft panels, wind turbine blades, and building cladding.

Questions & Answers (Continued)

Q 8

5 Marks

Explain the properties of refractories in detail.

The essential properties of a refractory material are:

- 1. Refractoriness (PCE):** The ability to withstand high temperatures without softening or melting under its own weight. Measured by PCE number. Higher PCE = greater refractoriness. Super duty refractories have $PCE > 33$ ($T > 1750^{\circ}\text{C}$).
- 2. Refractoriness Under Load (RUL):** Temperature at which the refractory deforms by 0.5% ($T_{0.5}$) under 0.2 N/mm^2 compressive load. Critical for structural stability in furnace roofs and arches.
- 3. Thermal Spalling Resistance:** Ability to resist cracking due to rapid temperature changes (thermal shock). Improved by low thermal expansion coefficient, high thermal conductivity, and high strength.
- 4. Chemical Resistance (Slag Resistance):** Resistance to chemical attack by molten slag — acidic refractories resist acidic slags (SiO_2 dominant); basic refractories resist basic slags (CaO , MgO dominant); neutral refractories resist both types.
- 5. Porosity and Permeability:** Low apparent porosity ($<15\text{--}20\%$) is preferred to prevent slag and gas penetration. High porosity leads to internal chemical attack and structural weakening.
- 6. Cold Crushing Strength (CCS):** Ability to bear load at room temperature during construction, transport, and initial heat-up. Ranges from 10 to 200 N/mm^2 .
- 7. Thermal Conductivity:** Should be low for heat conservation (furnace walls) but high for hearths requiring heat removal. Ranges from 0.5 to 30 W/mK .
- 8. Permanent Linear Change (PLC):** Change in dimensions after repeated heating. Should be $<\pm 2\%$ to prevent joint failure and structural collapse.

Explain the mechanism of lubrication with a comparison of thick film and thin film lubrication.

Lubrication is the process of reducing friction and wear between two surfaces in relative motion by introducing a lubricant. The three main mechanisms are:

1. Thick Film (Hydrodynamic) Lubrication: A continuous, unbroken film of lubricant completely separates the two surfaces. The relative motion of surfaces creates a convergent wedge of oil that generates hydrodynamic pressure sufficient to support the load — this is called the "wedge action." Characteristics: (i) No metal-to-metal contact, (ii) Friction coefficient is very low (0.001–0.01), (iii) Friction depends on viscosity of lubricant (governed by viscous shear of oil film), (iv) Operates at high sliding speeds and moderate loads, (v) Film thickness 10–100 μm . Examples: Journal bearings in engines, turbine bearings.

2. Thin Film (Boundary) Lubrication: Occurs at high loads or low speeds where the lubricant film breaks down to a few molecular layers. Some metal-to-metal contact occurs at surface asperities (microscopic peaks). Lubricant molecules are adsorbed (physically or chemically) onto the metal surfaces forming a protective monolayer. Characteristics: (i) Some asperity contact occurs, (ii) Friction coefficient moderate (0.05–0.15), (iii) Friction depends on chemical nature of lubricant, not viscosity, (iv) Operates at low speeds, high loads, and during start-stop cycles, (v) Film thickness 1–10 nm. Examples: Cutting operations, gear surfaces under high load.

3. Extreme Pressure (EP) Lubrication: Used when loads are too severe for boundary lubrication. EP lubricants contain reactive compounds (S, Cl, P) that react with metal surfaces at high asperity temperatures to form thin, low-shear-strength solid films (iron sulphide, chloride, phosphate), preventing seizure. Used in hypoid gear oils, rolling mills, cutting fluids.

The Stribeck curve shows transition between these regimes based on the bearing number ($\eta \times N / P$).

Explain the setting and hardening of Portland cement with hydration reactions.

When water is added to Portland cement, chemical reactions called hydration occur, leading to setting and hardening:

Hydration Reactions:

(1) $\text{C}_3\text{S} + \text{H}_2\text{O} \rightarrow \text{C-S-H Gel} + \text{Ca(OH)}_2 + \text{Heat}$ — Fast reaction, responsible for early (3–7 day) strength.

(2) $\text{C}_2\text{S} + \text{H}_2\text{O} \rightarrow \text{C-S-H Gel} + \text{Ca(OH)}_2$ — Slow reaction, responsible for long-term (28 day+) strength.

(3) $\text{C}_3\text{A} + \text{Gypsum} + \text{H}_2\text{O} \rightarrow \text{Ettringite } (3\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot 3\text{CaSO}_4 \cdot 32\text{H}_2\text{O})$ — Ettringite controls flash setting.

(4) $\text{C}_4\text{AF} + \text{H}_2\text{O} \rightarrow \text{Calcium aluminate hydrate} + \text{Iron oxide hydrate}$ — Moderate rate, contributes colour.

Setting Process: Initial setting (≥ 30 min) occurs when C_3A reacts with gypsum to form ettringite crystals that interlock and cause stiffening. Final setting (≤ 600 min) occurs when C_3S begins rapid hydration, forming C-S-H gel that binds paste into a solid mass.

Hardening Process: Over days to years, continued hydration of C_3S and C_2S produces more C-S-H gel — a dense, poorly crystalline material with specific surface area $150\text{--}700 \text{ m}^2/\text{g}$ — which fills pore spaces, reduces porosity, and dramatically increases compressive strength. At 28 days, approximately 70% of the ultimate strength is achieved. Curing (keeping concrete wet) is essential to maintain hydration and prevent premature drying. Without adequate curing, hydration stops and strength development ceases.

Calculate the Viscosity Index (VI) of a lubricating oil. The kinematic viscosity of the sample oil at 40°C is 75 cSt. The reference oil with VI = 0 has viscosity L = 120 cSt at 40°C. The reference oil with VI = 100 has viscosity H = 50 cSt at 40°C. (Both reference oils have the same viscosity as the sample at 100°C).

Given:

$$U = 75 \text{ cSt (sample at 40°C)}$$

$$L = 120 \text{ cSt (VI = 0 reference at 40°C)}$$

$$H = 50 \text{ cSt (VI = 100 reference at 40°C)}$$

Formula:
$$VI = (L - U) / (L - H) \times 100$$

Solution:

$$L - U = 120 - 75 = 45$$

$$L - H = 120 - 50 = 70$$

$$VI = (45 / 70) \times 100 = 0.6428 \times 100 = \mathbf{64.28 \approx 64}$$

Interpretation: VI = 64 means this oil has moderate viscosity–temperature characteristics. Since VI < 100, it is a single-grade oil. A VI > 100 would indicate a multigrade oil with excellent temperature performance. Adding VI improver additives (polymethacrylates) can increase the VI above 100.

Quick Revision Sheet

1. COMPOSITES – KEY POINTS

- Two phases: **Matrix** (continuous) + **Reinforcement** (dispersed)
- Types: **Particle** (large particle + dispersion) | **Fibre** (continuous, discontinuous, woven) | **Structural** (laminates + sandwich)
- Rule of Mixtures: $E_c = E_m V_m + E_f V_f$
- GFRP (glass), CFRP (carbon), KFRP (Kevlar) — FRC types
- Applications: Aerospace, automotive, marine, sports, civil

2. REFRACTORIES – KEY POINTS

- Definition: Materials stable above 1580°C
- Classification by chemistry: **Acidic** (SiO₂-rich, e.g., silica brick) | **Basic** (MgO/CaO-rich, e.g., magnesite) | **Neutral** (e.g., chromite, carbon)
- Classification by form: Shaped bricks | Unshaped (castables) | Refractory fibres
- Key properties: Refractoriness (PCE) | RUL | Thermal spalling resistance | Slag resistance | Porosity | CCS
- Thermal Shock Parameter: $R = (\sigma_f \times k) / (E \times \alpha)$

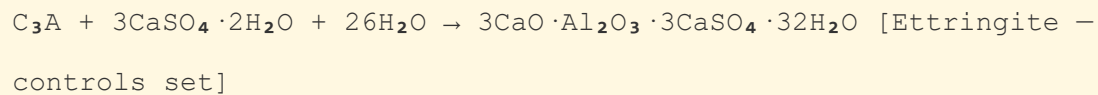
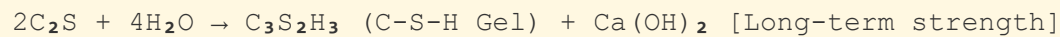
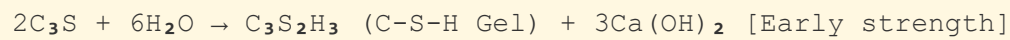
3. LUBRICANTS – KEY POINTS

- Functions: Reduce friction & wear, cooling, sealing, corrosion protection, cleaning
- Types: Liquid (oils) | Semi-solid (greases) | Solid (graphite, MoS₂, PTFE) | Gaseous (air bearings)
- Mechanisms: Thick film (hydrodynamic, $\mu \approx 0.001-0.01$) | Thin film / Boundary ($\mu \approx 0.05-0.15$) | EP
- **VI formula:** $VI = (L - U) / (L - H) \times 100$
- Flash Point: vapour ignites momentarily | Fire Point: burns for ≥ 5 seconds
- Cloud Point: wax separation | Pour Point: oil ceases to flow
- Saponification Value: mg KOH to saponify 1 g oil

4. PORTLAND CEMENT – KEY POINTS

- Bogue Compounds: C_3S (45–65%) | C_2S (15–30%) | C_3A (6–12%) | C_4AF (6–10%) | Gypsum (2–4%)
- C_3A = highest heat of hydration (867 J/g); controlled by gypsum → ettringite
- Setting: Initial ≥ 30 min; Final ≤ 600 min
- Key product of hydration = **C-S-H gel** (calcium silicate hydrate) — primary strength phase
- Hardening: continues for months — due to C_2S hydration

5. IMPORTANT REACTIONS – CEMENT HYDRATION



6. ALL KEY FORMULAS

$$\text{Viscosity Index: } VI = (L - U) / (L - H) \times 100$$

$$\text{Kinematic Viscosity: } \nu = \eta / \rho$$

$$\text{Dynamic Viscosity: } \eta = \text{Shear Stress} / \text{Shear Rate}$$

$$\text{Rule of Mixtures: } E_c = E_m \times V_m + E_f \times V_f$$

$$\text{Thermal Shock Resistance: } R = (\sigma_f \times k) / (E \times \alpha)$$

$$\text{Heat of Hydration (C}_3\text{A): 867 J/g | (C}_3\text{S): 502 J/g | (C}_2\text{S): 260 J/g}$$

7. LUBRICANT PROPERTIES – QUICK COMPARISON

Property	High Value Means	Low Value Means
Viscosity Index	Good temp. performance	Poor temp. performance
Flash Point	Safe, less volatile	Fire hazard
Cloud Point	Wax separates at higher temp	Better cold performance
Pour Point	Solidifies at higher temp	Flows at lower temp (better cold)
Saponification Value	More ester content	Purer mineral oil

Textbook References

Primary Textbooks

Jain P.C. and Jain M. – *Engineering Chemistry*, 16th Edition. Dhanpat Rai Publishing Company, New Delhi, 2013.

Primary Reference Comprehensive coverage of lubricants, refractories, composites, and cement chemistry for B.Tech level. Chapters 12–15 cover Unit IV topics in detail.

Atkins P., de Paula J., and Keeler J. – *Atkins' Physical Chemistry*, 10th Edition. Oxford University Press, 2010.

Primary Reference Provides thermodynamic basis for hydration reactions, viscosity theory, and surface chemistry relevant to lubricants.

Reference Books

Taylor H.F.W. – *Cement Chemistry*, 2nd Edition. Thomas Telford Publishing, London, 1997.

Cement Chemistry Definitive reference for Portland cement mineralogy, hydration kinetics, microstructure development, and C-S-H gel structure.

Shaw D.J. – *Introduction to Colloid and Surface Chemistry*, 4th Edition. Butterworth-Heinemann, Oxford, 1992.

Colloids & Surfaces Covers surface adsorption theory fundamental to boundary lubrication, surface chemistry of lubricants, and colloidal stability.

Billmeyer F.W. Jr. – *Textbook of Polymer Science*, 3rd Edition. John Wiley and Sons, New York, 1984.

Polymer Science Essential reference for polymer matrix composites — thermosets, thermoplastics, epoxy resins, and reinforced plastics.

Callister W.D. and Rethwisch D.G. – *Materials Science and Engineering: An Introduction*, 9th Edition. Wiley, 2014.

Additional Comprehensive coverage of composite materials (Chapter 16), ceramics as refractories, and materials selection.

Khanna O.P. – *A Textbook of Engineering Chemistry*. Dhanpat Rai Publications, New Delhi.

Additional Covers lubricants, refractories, and building materials in an exam-oriented format

Standards and Codes Referenced

IS: 269 – Specification for Ordinary, Rapid Hardening and Low Heat Portland Cement. Bureau of Indian Standards, New Delhi.

IS: 8112 – Specification for 43 Grade Ordinary Portland Cement. Bureau of Indian Standards.

ASTM C114 – Standard Test Methods for Chemical Analysis of Hydraulic Cement.

ASTM C27 – Standard Classification of Fireclay and High-Alumina Refractory Brick.

ASTM D2270 – Standard Practice for Calculating Viscosity Index from Kinematic Viscosity at 40°C and 100°C.

ASTM D92 – Standard Test Method for Flash and Fire Points by Cleveland Open Cup (COC) Tester.

Exam Strategy Note:

For university examinations, focus on: (1) Definitions with proper technical language, (2) Classification with examples in tabular form, (3) Reactions with correct formulae – use subscript notation carefully, (4) Diagrams – even rough schematic diagrams earn marks if properly labelled, (5) Numerical problems – show all steps with units. The Viscosity Index calculation is frequently asked as a 5-mark numerical in Anna University and similar board examinations.

End of Unit IV Module

This module covers the complete syllabus of Engineering Chemistry Unit IV – Modern Engineering Materials as per B.Tech First Year curriculum. All sections include theory, diagrams, formulae, reactions, and exam-oriented Q&A. For additional numerical practice, refer to solved examples in Jain & Jain (Chapter 15 – Lubricants and Chapter 13 – Cement).